Umesh Kashyap

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PROFILE SUMMARY

Ph.D. Scholar at the **Indian Institute of Technology (IIT) Bhilai** with 4+ years of research experience in **Trustworthy AI** and **Adversarial Machine Learning**. Specialized in developing robust defense mechanisms against adversarial attacks, perceptual encryption, secure AI inference, and deepfake detection and attribution.

Demonstrated track record of publishing in high-impact venues (**PETs**, **ICDAR**, **IEEE Transactions**) and securing competitive research funding (TIH, NFOBC). Seeking a Post-Doctoral position to leverage expertise in model security benchmarking, privacy-preserving AI, and forensic analysis to advance cutting-edge research in trustworthy AI safety.

EDUCATION

Ph.D. in Computer Science (Ongoing) *2022 - Present*
Indian Institute of Technology Bhilai, India
Thesis Focus: Adversarial Security Evaluation & Privacy-Preserving Deep Learning

M.Sc. Computer Science *2018 - 2020*
Atal Bihari Vajpayee University, Bilaspur, India

B.Sc. Computer Science *2015 - 2018*
Bilaspur University, Bilaspur, India

RESEARCH & PROFESSIONAL EXPERIENCE

PhD Research Scholar *Jan 2022 - Present*
MIST Lab, IIT Bhilai

- Led a security benchmark for perceptual encryption, exposing critical vulnerabilities in state-of-the-art methods.
- Designed "De-Fake," a style-based anomaly framework for deepfake detection that generalizes to unseen datasets.
- Evaluated malware detectors against evasion attacks and proposed robust countermeasures.

Teaching Assistant *2023 - Present*
Indian Institute of Technology (IIT) Bhilai

- **Load:** 8 hours/week.
- **Responsibilities:** Conducting tutorials, designing problem sets, preparing coding assignments, resolving student doubts, and evaluating assignments and examination papers.
- **Courses:**
  (a) Introduction to Programming (CSL-100), 2023-M, 2024-M.
  (b) Adversarial Machine Learning (CSL-607), 2024-W, 2025-W, 2026-W.
  (c) AI/ML Lab (DSP-301), 2026-M.

Project Scientist *Jan 2023 - Dec 2023*
Indian Institute of Technology (IIT) Delhi

- Evaluated next-generation wireless systems using ML-driven optimization.
- Validated performance metrics and prepared grant-compliance documentation.

Guest Faculty (Computer Science) *2020 - 2022*
Atal Bihari Vajpayee University

- Taught core curriculum: Machine Learning, Data Structures, and Programming.
- Mentored undergraduate students on capstone projects focusing on applied ML.

PUBLICATIONS

Published

1. S. K. Padhi, A. Tiwari, **U. Kashyap**, S. S. Ali, *"Doc-Protector: A Self-Healing Approach for Digital Documents."* **The 20th International Conference on Document Analysis and Recognition (ICDAR)**, 2026.
2. **U. Kashyap**, S. S. Ali, *"GAN-Invert: Unveiling Vulnerabilities in Privacy-Preserving Facial Transformations."* **Proceedings on Privacy Enhancing Technologies (PETs)**, 2026.

3. **U. Kashyap**, A. Vishwakarma, S. S. Ali, *"Generalize Adversarial Malware Detection."* **Journal of Hardware and Systems Security**, 10(1), p.11, 2026.
4. R. Kumar, **U. Kashyap**, S. S. Ali, *"Gradient-Guided Adversarial Patch Attack for Deep Neural Networks."* **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, pp. 227–245, 2026.
5. **U. Kashyap**, S. K. Padhi, S. S. Ali, *"Is Perceptual Encryption Secure? A Security Benchmark for Perceptual Encryption Methods."* **IEEE Transactions on Artificial Intelligence**, 2025.
6. A. Vishwakarma, **U. Kashyap**, S. S. Ali, *"Adversarial Malware Detection."* **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, pp. 277–286, 2024.

Under Publication

7. S. K. Padhi, D. Diwedi **U. Kashyap**, S. S. Ali, *"MIRROR: Makeup Identity Restoration for RObust Facial Recognition."* **International Conference on Computer Vision and Image Processing (CVIP)**, 2026. **Accepted**
8. **U. Kashyap**, R. Borikar, S. K. Padhi, S. S. Ali, *"The Cosine Conspiracy: Subverting CLIPure-Cos with a Latent-Space Informed Attack."* **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, 2026. **Accepted**
9. S. Nevase, **U. Kashyap**, S. S. Ali, *"Perceptually-Guided Adversarial Diffusion: Enhancing Visual Fidelity in Facial Privacy."* **International Conference on Computer Vision and Image Processing (CVIP)**, 2025. **Under Publication**

Preprints

10. S. K. Padhi, H. Kumar, **U. Kashyap**, S. S. Ali, *"De-Fake: Style-Based Anomaly Deepfake Detection."* preprint arXiv:2507.03334, 2025.
11. S. M. Shahid, S. K. Padhi, **U. Kashyap**, S. S. Ali, *"Generalized Deepfake Attribution."* preprint arXiv:2406.18278, 2024.

CONFERENCE PRESENTATIONS

- **Oral Presentation, PETs 2026 (Online):** GAN-Invert: Unveiling Vulnerabilities in Privacy-Preserving Facial Transformations
- **Oral Presentation, ICDAR 2025:** Doc-Protector: A Self-Healing Approach for Digital Documents
- **Oral Presentation, CVIP 2025:** Perceptually-Guided Adversarial Diffusion: Enhancing Visual Fidelity in Facial Privacy
- **Poster Presentation, IndoML 2025:** Is Perceptual Encryption Secure? A Security Benchmark for Perceptual Encryption Methods
- **Oral Presentation, SPACE 2024:** Adversarial Malware Detection

FELLOWSHIPS & AWARDS

- **TIH PhD Fellowship:** Awarded by the Technology Innovation Hub (TIH) for research on AI-driven GST fraud detection (Dec 2023 – May 2025).
- **National Eligibility Test (UGC-NET):** Qualified **6 times** (2019–2025), demonstrating consistent top-tier academic competency nationwide.
- **NFOBC Doctoral Fellowship:** Awarded by University Grants Commission (2023, 2024).
- **CG-SET Qualified:** Chhattisgarh State Eligibility Test (2019).
- **Travel Grants:**
  - **IndoML Symposium 2024 and 2025:** Travel grants to attend the symposium; presented a poster.
  - **CVIP 2025:** Travel grant to attend the conference and deliver an oral presentation on perceptually-guided adversarial diffusion for facial privacy.
  - **ICDAR 2025 (ANRF):** International travel grant to attend the conference and present the paper.
  - **PETs:** Travel grant to attend conference.

TECHNICAL SKILLS

Languages	Python, C, C++, LaTeX
Tools & Platforms	Git, Docker, Linux, Google Colab, Jupyter, Hugging Face
Core Competencies	Model Robustness, Adversarial Attack/Defense, Computer Vision

LANGUAGES

Hindi: Native **English:** Professional